

Video Transcript - Part 9

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Hey everyone, welcome to part nine. Today we're covering two data quality problems you'll run into constantly in real world machine learning, missing data, and unbalanced data. Neither one is glamorous, but how you handle them can make or break your model, often more than the choice of algorithm ever will.

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Here's the plan. We'll start with why data goes missing in the first place, then walk through techniques for handling it, and a quick guide for choosing between them. After that, we'll look at what unbalanced data actually is, why it quietly sabotages models, and the techniques and metrics you can use to deal with it.

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Let's start with missing data. There are plenty of reasons it shows up. Sensors fail and stop recording. People skip optional fields in a form. Datasets get merged and some records simply don't have matching values. And sometimes information is deliberately redacted for privacy. Whatever the cause, you almost never get a perfectly complete dataset, so you need a plan for dealing with the gaps.

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There are a few common ways to handle missing data. The simplest is deletion, just dropping rows or columns with missing values, though this can throw away useful data if it happens too often. Mean, median, or mode imputation fills gaps with the average or most common value, quick and easy, though it can distort the original distribution. Forward or backward fill is common for time series data, carrying the last known value forward to fill a gap. And predictive imputation goes a step further, using a model like regression or k nearest neighbors to estimate what the missing value probably would have been, based on the other features.

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So how do you actually choose between these? As a quick guide, if only a few values are missing with no clear pattern, deletion is usually fine. If it's a numeric column and the values seem to be missing at random, mean or median imputation works well. For time series data, forward or backward fill is the natural choice. And if the fact that a value is missing actually relates to other features in your data, that's a sign you should reach for predictive imputation instead.

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Now let's shift to unbalanced data. This shows up in classification problems, where one class massively outnumbers another. The classic example is fraud detection, where maybe ninety nine percent of transactions are legitimate, and only one percent are fraudulent. You'll see the same pattern in things like disease screening or equipment failure prediction. That kind of imbalance is extremely common, and it creates a subtle but serious problem.

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Here's the problem, in plain terms. If a model just predicts "not fraud" for every single transaction, it would be right ninety nine percent of the time, and look highly accurate. But it would never catch a single actual fraud case, which is exactly the thing you built the model to catch in the first place. This is sometimes called the accuracy trap, high accuracy can hide a completely useless model when your classes are imbalanced.

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So how do you deal with this? Resampling is the most common approach, and it comes in two flavors. Oversampling adds more examples of the minority class, sometimes by duplicating them, or generating realistic synthetic examples using a technique called SMOTE.

Undersampling goes the other way, trimming down the majority class so the two are more balanced. Another approach is class weighting, where you tell the model to penalize mistakes on the minority class more heavily during training, so it pays closer attention to it. And when it's feasible, simply collecting more real examples of the minority class helps too.

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And just as important as fixing the data is fixing how you measure success. Accuracy alone is misleading for imbalanced problems, so instead, look at precision, how many of your positive predictions were actually correct, recall, how many of the actual positives you managed to catch, the F1 score, which balances the two, or AUC-ROC, which measures how well the model separates the classes across every possible threshold.

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So to recap, missing data can be handled through deletion, simple imputation, or predictive methods, depending on how much you can afford to lose or estimate. Unbalanced data needs its own toolkit, resampling, class weighting, and crucially, evaluation metrics that actually reflect performance on the class you care about. Get both of these right, and your model will be built on a much more honest foundation. Thanks for watching, and I'll see you in the next one.